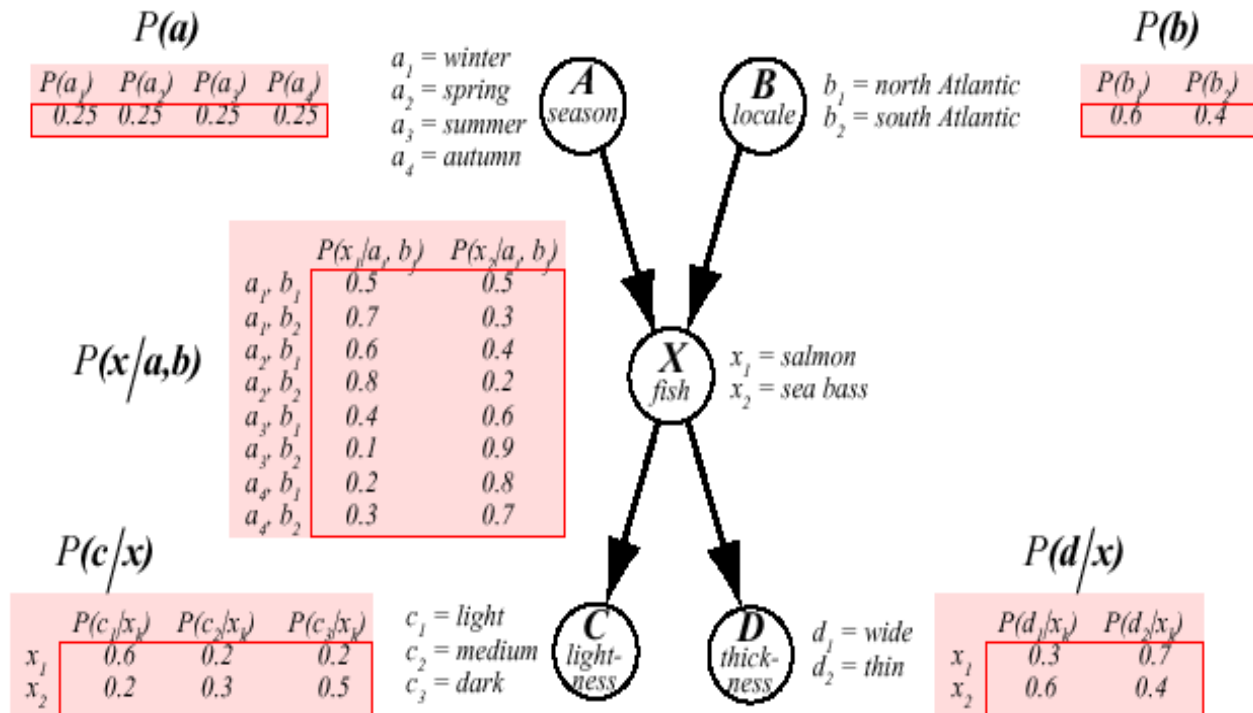




i) We can compute the probability of any configuration of states in the joint density, e.g.:

$$P(a_3, b_1, x_2, c_3, d_2) = P(a_3)P(b_1)P(x_2/a_3, b_1)P(c_3/x_2)P(d_2/x_2) =$$

$$0.25 \times 0.6 \times 0.4 \times 0.5 \times 0.4 = 0.012$$

- In general, if X denotes the query variables and e denotes the evidence, then

$$P(X/e) = \frac{P(X, e)}{P(e)} = \alpha P(X, e)$$

Where $\alpha = 1/P(e)$ is a constant of proportionality.

ii) Classify a fish given that the fish is light (c_1) and was caught in south Atlantic (b_2) -- no evidence about what time of the year the fish was caught nor its thickness.

$$\begin{aligned}
P(x_1|c_1, b_2) &= \frac{P(x_1, c_1, b_2)}{P(c_1, b_2)} \\
&= \alpha \sum_{\mathbf{a}, \mathbf{d}} P(x_1, \mathbf{a}, b_2, c_1, \mathbf{d}) \\
&= \alpha \sum_{\mathbf{a}, \mathbf{d}} P(\mathbf{a}) P(b_2) P(x_1|\mathbf{a}, b_2) P(c_1|x_1) P(\mathbf{d}|x_1) \\
&= \alpha P(b_2) P(c_1|x_1) \\
&\quad \times \left[\sum_{\mathbf{a}} P(\mathbf{a}) P(x_1|\mathbf{a}, b_2) \right] \left[\sum_{\mathbf{d}} P(\mathbf{d}|x_1) \right]
\end{aligned}$$

$$\begin{aligned}
&= \alpha P(b_2) P(c_1|x_1) \\
&\quad \times [P(a_1)P(x_1|a_1, b_2) + P(a_2)P(x_1|a_2, b_2) + P(a_3)P(x_1|a_3, b_2) + P(a_4)P(x_1|a_4, b_2)] \\
&\quad \times \underbrace{[P(d_1|x_1) + P(d_2|x_1)]}_{=1} \\
&= \alpha (0.4)(0.6) [(0.25)(0.7) + (0.25)(0.8) + (0.25)(0.1) + (0.25)(0.3)] 1.0 \\
&= \alpha 0.114.
\end{aligned}$$

- Similarly,

$$P(x_2|c_1, b_2) = \alpha 0.066$$

- Normalize probabilities (not needed necessarily):

$$P(x_1|c_1, b_2) + P(x_2|c_1, b_2) = 1 \quad (\alpha = 1/0.18)$$

- $P(x_1|c_1, b_2) = 0.73$
 - $P(x_2|c_1, b_2) = 0.27$
- } Answer is Salmon

Lecture 6

Bayesian Belief Networks Example 1

Example: A Lecturer's Life

$P(io) = 0.6$
 $P(lo | io) = 0.5$
 $P(lo | \neg io) = 0.05$
 $P(co | io) = 0.8$
 $P(co | \neg io) = 0.1$

$p(A, B, C) = p(B|A)p(C|A)p(A)$

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Lecture 6

Bayesian Belief Networks Example 1

Example: A Lecturer's Life

$$P(io | co) = \frac{P(co | io)P(io)}{P(co)} = \frac{0.8(0.6)}{0.52} = 0.923$$

$$P(co) = P(co | io)P(io) + P(co | \neg io)P(\neg io)$$

$$= 0.8(0.6) + 0.1(0.4)$$

$$= 0.52$$

☐ You can also calculate posterior Probability $P(io | lo)$

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Bayesian Belief Networks Example 1

Example: A Lecturer's Life

$$\begin{aligned}
 P(lo|co) &= \frac{P(lo,co)}{P(co)} \\
 &= \frac{P(lo,co|io)P(io) + P(lo,co|\neg io)P(\neg io)}{P(co)} \\
 &= \frac{P(lo|io)P(co|io)P(io) + P(lo|\neg io)P(co|\neg io)P(\neg io)}{P(co)} \\
 &= \frac{.5(.8)(.6) + .05(.1)(.4)}{.52} = .465
 \end{aligned}$$

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Bayesian Belief Networks Example 2

let us consider the following scenario. Lakshman travels by air if he is on an official visit. If he is on a personal visit, he travels by air if he has money. If he does not travel by plane, he travels by train but sometimes also takes a bus. Draw BBN and calculate the probability that Lakshman travels by train, does not travel by air, goes on an official trip and has money using the give probability tables.

Solution

The variables involved are :

- 1. Lakshman travels by air (A)
- 2. Goes on official visit (F)
- 3. Lakshman has money (M)
- 4. Lakshman travels by train (T)
- 5. Lakshman travels by bus (B)

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Bayesian Belief Networks Example 2

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Bayesian Belief Networks(Lec6)

Bayesian Belief Networks Example 2

- ☐ In the graph, we can see the dependencies with respect to the variables.
- ☐ The probability values at a variable are dependent on the value of its parents. In this case, the variable A is dependent on F and M.
- ☐ The variable T is dependent on A and variable B is dependent on A.
- ☐ The variables F and M are independent variables which do not have any parent node. So their probabilities are not dependent on any other variable.
- ☐ Node A has the biggest conditional probability table as A depends on F and M. T and B depend on A.

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Bayesian Belief Networks(Lec6)

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Bayesian Belief Networks Example 2

- Once the graph is drawn which is a directed acyclic graph (DAG), the probabilities of the variables depending on the values of their parent node has to be entered.
- This requires us to know the problem at hand and estimate these probabilities. So for each node the Conditional Probability Table has to be entered.
- First we take the independent nodes.
 - Node F has a probability of $P(F) = 0.7$.
 - Node M has a probability of $P(M) = 0.3$.

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Bayesian Belief Networks(Lec6)

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Bayesian Belief Networks Example 2

- We next come to node A. The conditional probability table for this node can be represented as

F	M	$P(A F, M)$
T	T	0.98
T	F	0.98
F	T	0.98
F	F	0.10

- The conditional probability table for T can be represented as:

A	$T A$
T	0.0
F	0.6

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Bayesian Belief Networks Example 2

□ The conditional probability table for B is:

A	B A
T	0.0
F	0.40

□ Using the bayesian belief network, we can get the probability that Lakshman travels by **train, does not travel by air, goes on an official trip and has money**. In other words, we are finding $P(T, \neg A, F, M)$. The probability of each variable given its parent is found and multiplied together to give the probability.

□ $P(T, \neg A, F, M) = P(T | \neg A) * P(\neg A | F \text{ and } M) * P(F) * P(M)$
 $= 0.6 * 0.98 * 0.7 * 0.3 = 0.123$

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